

SARS-CoV-2 NSP13 interacts with TEAD to suppress Hippo-YAP signaling

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eLife Assessment

This **important** study elucidates the molecular function of the SARS-CoV-2 helicase NSP13, which inhibits the transcriptional activity of the YAP/TEAD complex in vitro and in vivo. The evidence supporting the authors' claims is **compelling**, based on cell biological assays and multi-omic studies. This work contributes to the understanding of the new regulatory mechanism of YAP/TEAD after SARS-CoV-2 infection and will be of interest to researchers investigating COVID-19 infection and the Hippo-YAP signaling pathway.

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Abstract

The Hippo pathway controls organ development, homeostasis, and regeneration primarily by modulating YAP/TEAD-mediated gene expression. Although emerging studies report Hippo-YAP dysfunction after viral infection, it is largely unknown in the context of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Here, we analyzed RNA sequencing data from induced pluripotent stem cell-derived cardiomyocytes (iPSC-CMs) and SARS-CoV-2-infected human lung samples, and observed a decrease in YAP target gene expression. In screening SARS-CoV-2 nonstructural proteins, we found that nonstructural protein 13 (NSP13), a conserved coronavirus helicase, inhibits YAP transcriptional activity independent of the upstream Hippo kinases LATS1/2. Consistently, introducing NSP13 into cardiomyocytes suppresses an active form of YAP (YAP5SA) *in vivo*. Subsequent investigations on NSP13 mutants revealed that NSP13 helicase activity, including DNA binding and unwinding, is crucial for suppressing YAP transactivation. Mechanistically, TEAD4 serves as a platform to recruit NSP13 and YAP. NSP13 likely inactivates the YAP/TEAD4 transcription complex by remodeling chromatin to recruit proteins, such as transcription termination factor 2 (TTF2), to bind the YAP/TEAD/NSP13 complex. These findings reveal a novel YAP/TEAD regulatory mechanism and uncover molecular insights into Hippo-YAP regulation after SARS-CoV-2 infection.

Introduction

The evolutionarily conserved Hippo-YAP signaling pathway integrates various extracellular signals, including mechanical force, cell adhesion, and nutrient availability, through the protein kinases LATS1/2 (1 [↗](#)). YAP is the primary substrate of LATS kinases, and following phosphorylation, is inactivated via cytoplasmic retention or degradation. Unphosphorylated YAP translocates to the nucleus and complexes with transcription factor partners, notably TEAD, to activate target genes (1 [↗](#), 2 [↗](#)) involved in tissue development, homeostasis, and regeneration across multiple organs (2 [↗](#)–7 [↗](#)). YAP/TEAD is regulated by intricate molecular mechanisms that govern its activity, localization, and interaction. Dysregulation of these processes has been implicated in various diseases, including cancer (8 [↗](#)–11 [↗](#)).

Recent studies reveal that YAP is an endogenous brake of innate antiviral immunity (12 [↗](#), 13 [↗](#)). Genetic loss of YAP enhances antiviral responses, whereas expression of a transactivation deficient mutant (YAP 6SA) suppresses these responses. Conversely, multiple canonical YAP target genes, such as *IL-6*, *Ccl2*, and *Csf1*, modulate innate immune signaling and cytokine production (14 [↗](#)–16 [↗](#)). Notably, different types of viral infection distinctly influence YAP expression, degradation, and nuclear localization (17 [↗](#)). Although these observations shed light on the complex dynamics of YAP during viral responses, the mechanisms underlying YAP/TEAD regulation after SARS-CoV-2 infection are poorly understood. SARS-CoV-2, the center of the COVID-19 pandemic, has well-documented effects on the respiratory, digestive, central nervous, and cardiovascular systems (18 [↗](#)–20 [↗](#)). Investigating SARS-CoV-2 and Hippo-YAP signaling provides molecular insight into virus-induced pathophysiology. Two groups recently reported opposite effects on YAP activity after SARS-CoV-2 infection (21 [↗](#), 22 [↗](#)). However, the lack of YAP/TEAD target gene expression data in these studies prevents a full understanding of these disparate results. Thus, the precise understanding of YAP/TEAD regulation after SARS-CoV-2 infection remains elusive.

Here, we found that SARS-CoV-2 infection reduces YAP target gene expression in human induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs) and lung epithelial cells of COVID-19 patients. By comprehensively screening SARS-CoV-2 nonstructural proteins, we identified NSP13 as a key factor in inhibiting YAP transcriptional activity and suppressing active YAP5SA activity *in vivo*. Moreover, NSP13 suppression of YAP is dependent on its helicase activity, where both DNA binding and unwinding are required. Mechanistically, NSP13 is directly bound to TEAD4 and inhibits the transcriptional activity of the YAP/TEAD4 complex by remodeling chromatin and recruiting transcriptional suppressors such as TTF2. These findings reveal a novel function of NSP13 in regulating YAP/TEAD activity and provide key insights into how the SARS-CoV-2 genome modulates transcriptional activity of the YAP-TEAD complex.

Results

SARS-CoV-2 infection suppresses YAP activity in host cells

To assess YAP activity following SARS-CoV-2 infection *in vitro*, we evaluated a bulk RNA-sequencing dataset derived from hiPSC-CMs exposed to varying SARS-CoV-2 concentrations (23 [↗](#)) (Figure 1A [↗](#)). Cardiomyocyte-specific YAP target gene (24 [↗](#)) expression revealed that YAP targets were decreased in a dose-dependent manner after SARS-CoV-2 infection (Figure 1B [↗](#)). YAP is known to directly regulate several innate immunity genes, including *Ccl2*, *Thbs1*, and *Csf1* (15 [↗](#), 16 [↗](#)). We observed that these genes are down-regulated in SARS-CoV-2 infected cells, mirroring the decrease in canonical YAP targets such as *Vgll3*, *Amotl2*, and *Ccn1* (Figure 1C [↗](#)).

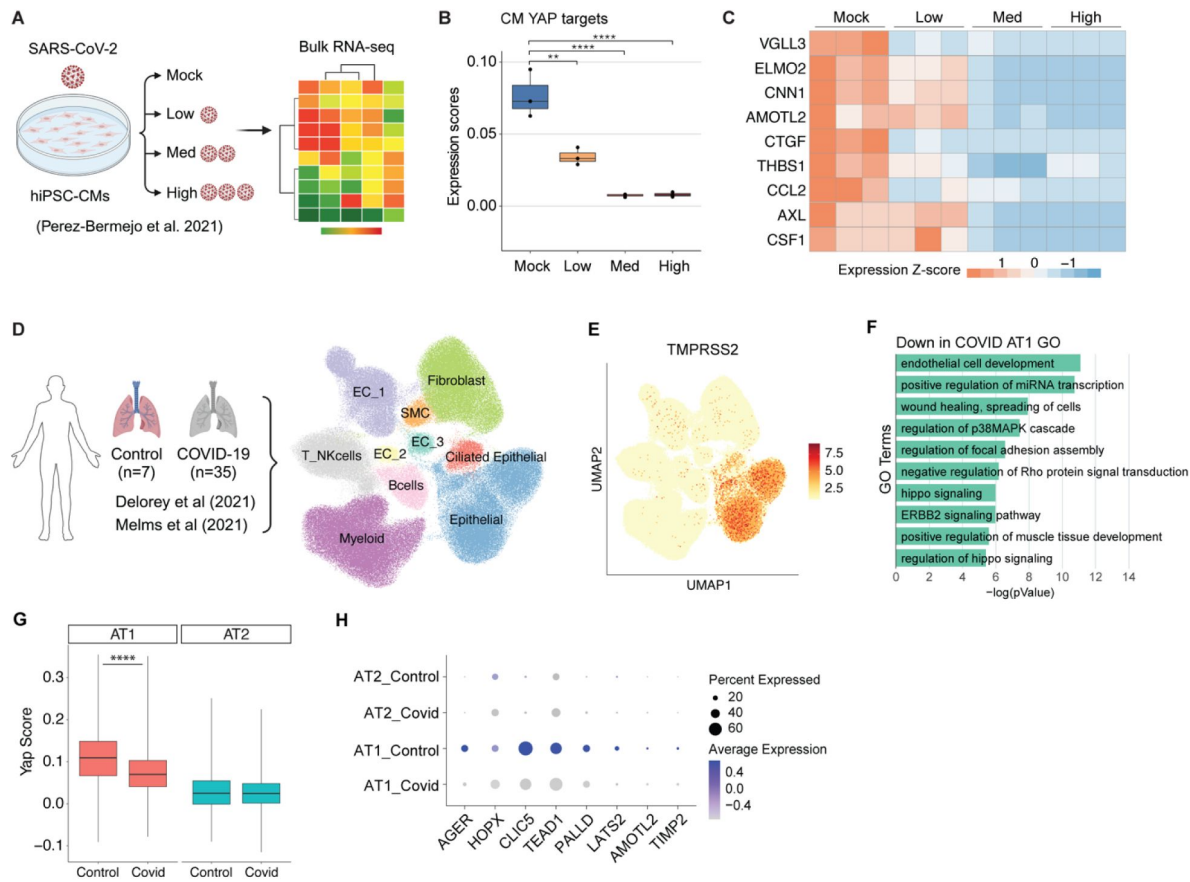


Figure 1.

SARS-CoV-2 infection suppresses YAP activity.

(A) Overview of SARS-CoV-2 infection in human induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs). (B) Box plot showing the mean expression scores of known YAP target genes in hiPSC-CM bulk RNA-sequencing data. Each dot represents a biological replicate. Student *t*-test; ***p* < 0.01, *****p* < 0.0001. (C) Heatmap displaying the expression Z-scores of example YAP targets in the iPSC-CM bulk RNA-sequencing data. Each column corresponds to a single biological sample. (D). Overview of integrated single-nucleus RNA sequencing and uniform manifold approximation and projection (UMAP) of cell types in lung samples from controls and patients with COVID-19. EC, endothelial cells; NK, natural killer cells; SMC, smooth muscle cells. (E) UMAP visualization of TMPRSS2 expression in the 10 cell types. (F) GO analysis of downregulated genes in AT1 cells from COVID patients. (G) YAP scores in alveolar type 1 (AT1) and alveolar type 2 (AT2) epithelial cells from lung samples in controls and patients with COVID-19. Wilcoxon test, *****p* < 0.0001. (H) Expression of YAP targets in AT1 and AT2 cells of control and COVID-positive lung samples.

We then assessed YAP activity with SARS-CoV-2 infection *in vivo* by integrating and reanalyzing single-nuclei RNA (snRNA) sequencing data derived from human lung samples (25, 26). We identified 10 major cell types (Figure 1D, Figure Supplement 1A). TMPRSS2 and ACE2, two key entry factors for SARS-CoV-2 infection (27) were more highly expressed in lung epithelial cells compared to other cell types (Figure 1E, Figure Supplement 1B). Lung epithelial cells were categorized into alveolar type 1 (AT1) and alveolar type 2 (AT2) (Figure Supplement 1C, D). SARS-CoV-2 is more likely to infect AT1 cells as they cover >95% of the alveolar surface. We performed an unbiased GO analysis of the differentially expressed genes (DEGs) and observed that genes associated with or regulating the Hippo pathway were downregulated in AT1 cells from COVID-19 patients (Figure 1F). Consistent with previous data *in vitro*, the YAP score, evaluated via YAP target gene expression, was lower in AT1 cells from COVID-19 patients than in controls (Figure 1G). Signature genes associated with AT1 cells, including the reported YAP target genes *AGER* and *CLIC5* (28, 29) were downregulated in COVID-19 patient lung samples (Figure 1H). These data support the notion that SARS-CoV-2 infection suppresses YAP activity in host cells. However, further analysis revealed that multiple YAP target genes involved in innate immunity and cytokine signaling were paradoxically elevated (Figure Supplement 1E). This discrepancy likely reflects a combination of factors, including cell type specificity, activation of parallel signaling pathways, and alterations in mechanical cues or tissue architecture, that can independently drive expression of these genes.

NSP13 inhibits YAP5SA transactivation

To identify which SARS-CoV-2 protein suppresses YAP activity, we screened 11 NSP proteins using a dual-luciferase reporter assay. This assay measures firefly and Renilla luciferase activities sequentially from the same sample. Normalization to Renilla luciferase serves as an internal control, ensuring accuracy and reproducibility of the results. Compared with other NSPs, NSP13 strongly inhibited YAP transcription activity (Figure 2A). By using a constitutively active form of YAP (YAP5SA) that is resistant to phosphorylation and inactivation by LATS1/2 (30), we observed that NSP13 suppresses YAP transactivation independent of its upstream kinase LATS2 (Figure Supplement 2A). Moreover, NSP13 attenuated YAP5SA activity dose-dependently (Figure 2B). A recent study reported that NSP13 suppresses episomal DNA transcription, as evidenced by reduced Renilla luciferase activity and decreased GFP expression upon co-expression with NSP13 (31). To evaluate this in our system, we examined our raw Renilla luciferase data and found that while 100 ng of NSP13 had no effect, 400 ng of NSP13 reduced Renilla luciferase levels by approximately 50% compared to the YAP5SA-only group (Figure Supplement 2B and 3C-D). Notably, the firefly luciferase signal, driven by YAP/TEAD interaction (HOP-Flash), exhibited an even greater reduction. To assess the specificity of this suppression, we also conducted a Notch reporter assay and observed that co-expression of NSP13 with NICD (Notch intracellular domain) did not inhibit Notch signaling (Figure Supplement 2C). These results reveal a specific suppressive effect of NSP13 on YAP-mediated transcriptional activity.

To investigate NSP13 function *in vivo*, we used YAP5SA transgenic mice (aMyHC-MerCreMer;YAP5SA), which express YAP5SA in cardiomyocytes following tamoxifen administration. YAP5SA induction produces cardiomyocyte hyperplasia and increased ejection fraction (EF) and fractional shortening (FS), ultimately leading to mortality (24). We used the adenovirus expressing NSP13 (AAV9-NSP13), which specifically infects cardiomyocytes (Figure Supplement 2D and Figure 2C). NSP13-induced expression in YAP5SA mouse cardiomyocytes increased survival rates (Figure 2D) and restored cardiac function (Figure 2E and Figure Supplement 2E). Furthermore, NSP13 expression reversed the smaller left ventricle (LV) chamber in YAP5SA mouse hearts (24) (Figure 2F, G). To investigate NSP13 induction in the mouse heart further, we collected heart tissue from surviving mice 21 days after tamoxifen injection. Notably, NSP13 expression in YAP5SA mice reversed heart overgrowth (Figure 2H, I, and Figure Supplement 2F). Together, these data indicate that NSP13 suppresses YAP activity *in vitro* and *in vivo*.

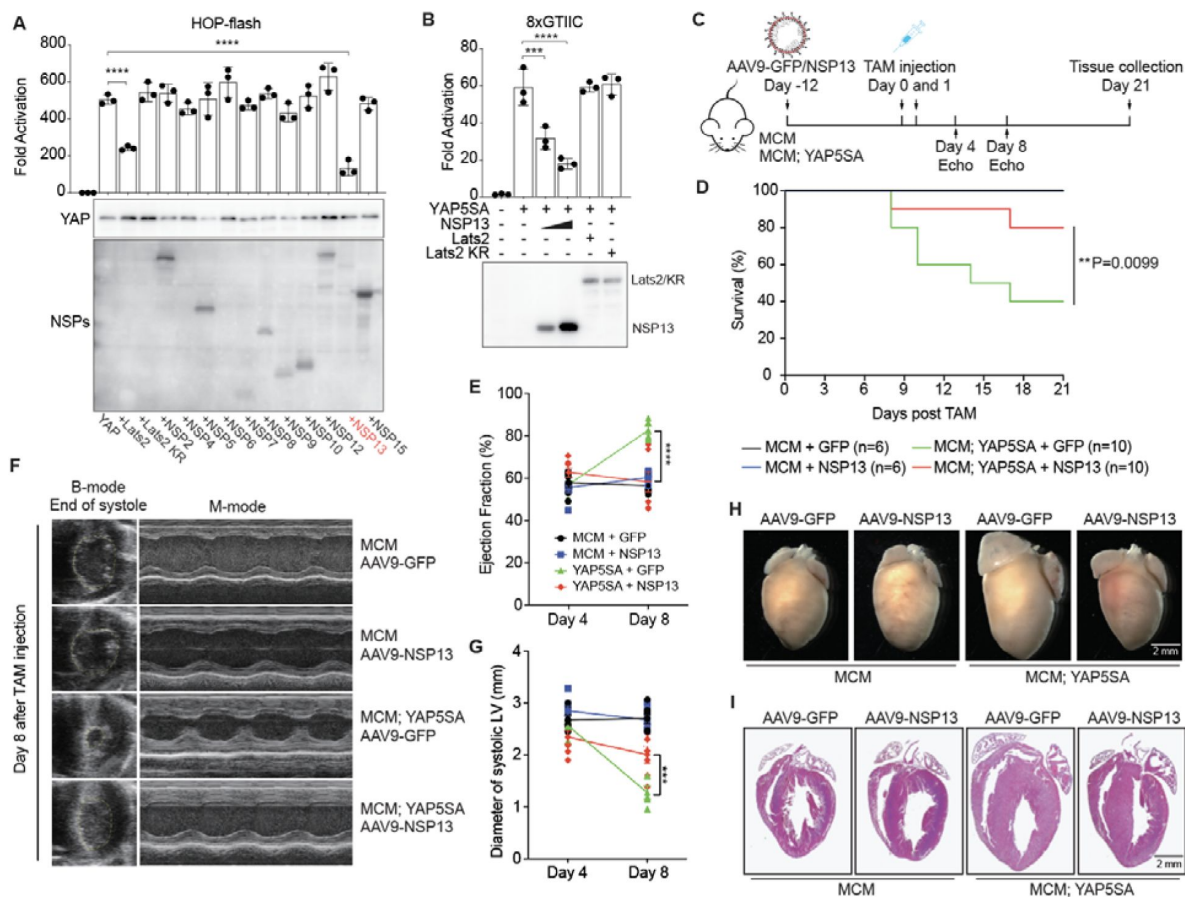


Figure 2.

NSP13 inhibits YAP transactivation.

(A) Screening of 11 NSPs for YAP activation by using a dual-luciferase reporter assay (HOP-flash). Compared with other NSPs, NSP13 strongly inhibited YAP transactivation at low protein expression levels. ($n = 3$ independent experiments; data are reported as mean $\pm$ SD). **** $p < 0.0001$, one-way ANOVA. (B) Reporter assay (8xGTIIC) results showing that NSP13 but not YAP upstream kinase LATS2 inhibited YAP5SA transactivation in a dose-dependent manner. ($n = 3$ independent experiments; data are presented as the mean $\pm$ SD). *** $p < 0.001$, **** $p < 0.0001$, one-way ANOVA. (C) Experimental design of NSP13 study in mice. Control (aMyHC-MerCreMer;WT) and YAP5SA (aMyHC-MerCreMer;YAP 5SA) mice were injected with AAV9-GFP or AAV9-NSP13. At 12 days after virus injection, the mice received two low doses of TAM (10 μ g/g). Cardiac function was recorded by echocardiography at day 4 and 8 after the second shot of tamoxifen. Hearts of all surviving mice were collected at day 21 post-tamoxifen injection. (D) NSP13 expression in cardiomyocytes improved the survival rate of YAP5SA mice after TAM injection compared to YAP5SA mice with AAV9-GFP infection. ** $p = 0.0099$, log-rank (Mantel-Cox) test. (E) Ejection fraction in YAP5SA mice was increased on day 8 after tamoxifen injections (10 μ g/g x2). NSP13 expression reversed the increase of EF in YAP5SA mice. **** $p < 0.0001$, three-way ANOVA. (F) Representative B-mode and M-mode echocardiographic images of mouse hearts in four groups, 8 days after tamoxifen (TAM) induction. (G) A reduction in the left ventricle size was seen in YAP5SA mice at day 8 after tamoxifen injection. NSP13 introduction reversed this trend as evidenced by an increase in the diameter of the left ventricle. *** $p < 0.001$, three-way ANOVA. (H-I) Representative whole mount and hematoxylin & eosin images of mouse hearts at 21 days after tamoxifen induction. Scale bar, 2 mm.

NSP13 helicase activity is required for YAP suppression

NSP13, a helicase with conserved sequence across all coronaviruses (**Figure 3A**), plays a critical role in viral replication and is a promising target for antiviral treatment (32–39). K131, K345/K347, and R567 were identified as important amino acid sites for NSP13 helicase activity in SARS-CoV (40). Given the 99.8% sequence identity of NSP13 between SARS-CoV-2 and SARS-CoV, we speculated that these sites were similarly crucial for SARS-CoV-2 NSP13. Reporter assays revealed that NSP13-K131A, which retains partial helicase activity, can suppress YAP, whereas NSP13-R567A (no ATP consumption) and NSP13-K345A/K347A (obstructed nucleic acid binding channel) failed to inhibit YAP transcriptional activity (**Figure 3B**). To assess the contribution of each domain to YAP regulation, we constructed six NSP13 truncations (**Figure 3C**), yet none of these truncations reduced YAP transactivation (**Figure 3D**). These findings suggest that full-length NSP13 with helicase activity (DNA binding and ATP dependent unwinding) is required to suppress YAP transactivation, and that partial helicase function is sufficient to fully inhibit YAP.

SARS-CoV-2 has consistently mutated over time, resulting in variants that differ from the original virus. We evaluated NSP13 mutations (35) (**Figure 3E** and **Figure Supplement 3A**) and observed that all mutants can suppress YAP transactivation (**Figure 3F** and **Figure Supplement 3B**).

NSP13 interacts with TEAD4 and recruits YAP repressors

To determine the molecular mechanism(s) underlying NSP13 suppression of YAP, we performed immunofluorescence (IF) analyses. Our data revealed that NSP13 localizes to the cytoplasm and nucleus, whereas most YAP5SA colocalizes with NSP13 in the nucleus three days after tamoxifen injection (**Figure 4A**). However, co-immunoprecipitation (co-IP) assays revealed no direct interaction between NSP13 and YAP5SA (**Figure Supplement 4A**). Additional assays indicated that NSP13 is associated with the transcription factor TEAD4 (**Figure 4B**), and both the N- and C-terminal ends of TEAD4 interact with NSP13 (**Figure Supplement 4B**), suggesting that NSP13 prevents YAP transactivation by competitively interacting with TEAD4. Unexpectedly, introduction of NSP13 had no effect on the YAP/TEAD4 association (**Figure Supplement 4C**). However, the YAP/NSP13 interaction was much stronger in the presence of TEAD4, indicating that TEAD4 is a platform to recruit YAP and NSP13 (**Figure 4C**). Consistent with this, IF analysis of HeLa cells and heart sections revealed that YAP5SA is restricted to the nucleus when co-expressed with either NSP13 WT or R567A, in contrast to the vector control (**Figure 4D**, **Figure Supplement 4D**). Moreover, NSP13 protein levels in cardiomyocytes accumulated following YAP5SA induction (**Figure 4D**, E), suggesting that nuclear NSP13/YAP/TEAD4 prevents NSP13 degradation. Given that NSP13 preferentially interacts with TEAD4 in the nucleus (**Figure 4B-C**, **Figure Supplement 4C**), we next determined whether this association was DNA dependent using multiple nucleases: Universal Nuclease (which degrades all forms of DNA and RNA), DNase I (which cleaves both single and double stranded DNA), and RNase H (which selectively cleaves the RNA strand in RNA/DNA hybrids). These treatments did not disrupt the NSP13/TEAD4 interaction (**Figure Supplement 4E**), revealing that their binding is not dependent on nucleic acids.

Given that NSP13 forms a complex with YAP and TEAD4, we hypothesized that it recruits nuclear repressors to inhibit YAP transactivation. To investigate this, we performed immunoprecipitation followed by mass spectrometry (IP-MS) to identify NSP13-interacting proteins in the presence or absence of YAP in HEK293T cell nuclei (**Figure 4F**). Our analyses identified hundreds of candidate proteins that interact with NSP13 in the nucleus (**Figure 4G**), which were further analyzed by STRING-based network analysis (**Figure Supplement 5A**). Gene Ontology (GO) analysis indicated that the largest clusters are involved in RNA polymerase II transcription termination, chromatin remodeling, and protein folding (**Figure 4H**). To validate these candidate interactors, we evaluated several proteins—CCT3, SMARCD1, EIF4A1, LMNA, TTF2, and YY2—and conducted co-IP assays. Whereas TEAD4 strongly interacts with NSP13 (**Figure**

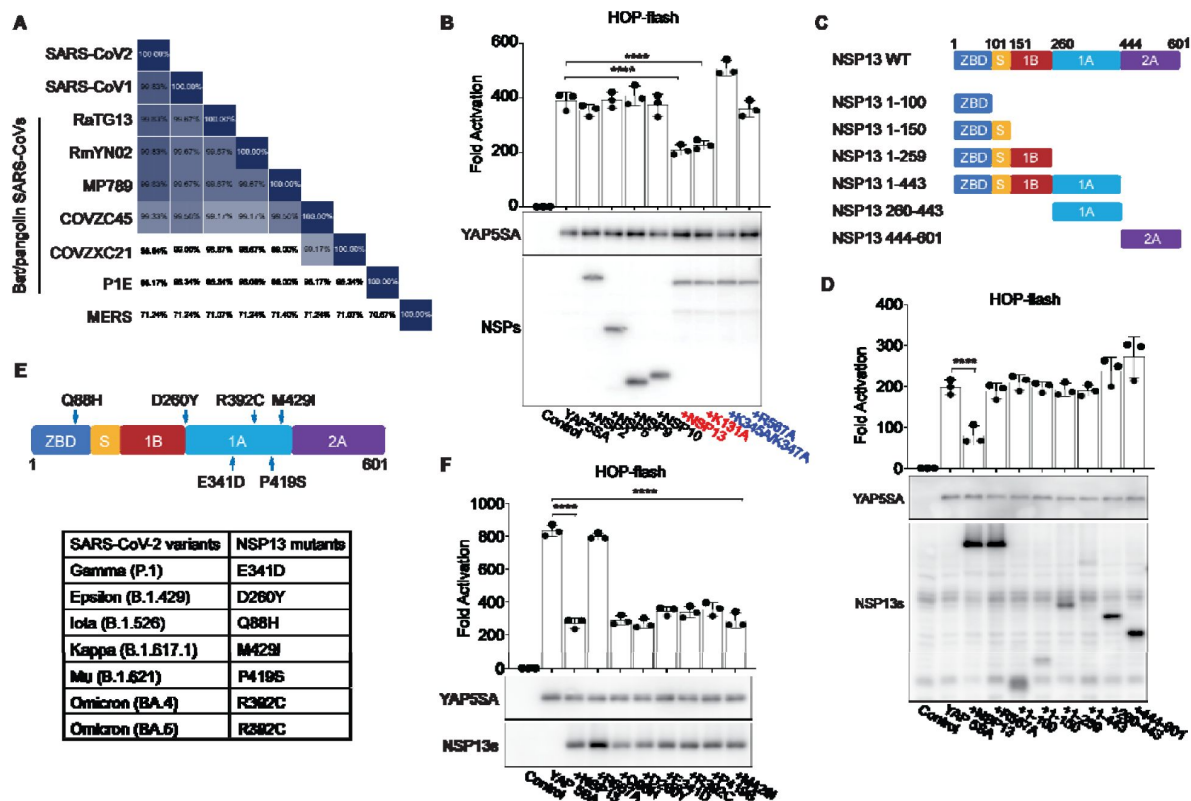


Figure 3.

NSP13 helicase activity is required for suppressing YAP activity.

(A) Conserved amino acid sequences of NSP13 among coronaviruses. (B) SARS-CoV-2 NSP13 mutant plasmids were constructed to examine YAP suppression mechanisms. NSP13-R567A, which loses its ATP consumption ability, did not inhibit YAP5SA transactivation, whereas NSP13 K345A/K347A, which loses its nucleic acid binding activity, mildly promoted YAP5SA transactivation. ($n = 3$ independent experiments; data are reported as the mean $\pm$ SD). $**p < 0.01$, $****p < 0.0001$, one-way ANOVA. (C) 6 NSP13 truncations were constructed based on the NSP13 domain map. (D) Reporter assay: none of the truncations led to a reduction in YAP transactivation, and the NSP13 DNA binding domains 1A and 2A slightly increased YAP5SA activation, suggesting that the full-length NSP13 with helicase activity may be required for suppression of YAP transactivation. ($n = 3$ independent experiments; data are reported as the mean $\pm$ SD). $*p < 0.05$, $****p < 0.0001$, one-way ANOVA. (E) Summary of NSP13 mutants from SARS-CoV-2 variants. (F) HOP-flash reporter assay: NSP13 mutations did not affect suppression of YAP5SA transactivation. $****p < 0.0001$, one-way ANOVA.

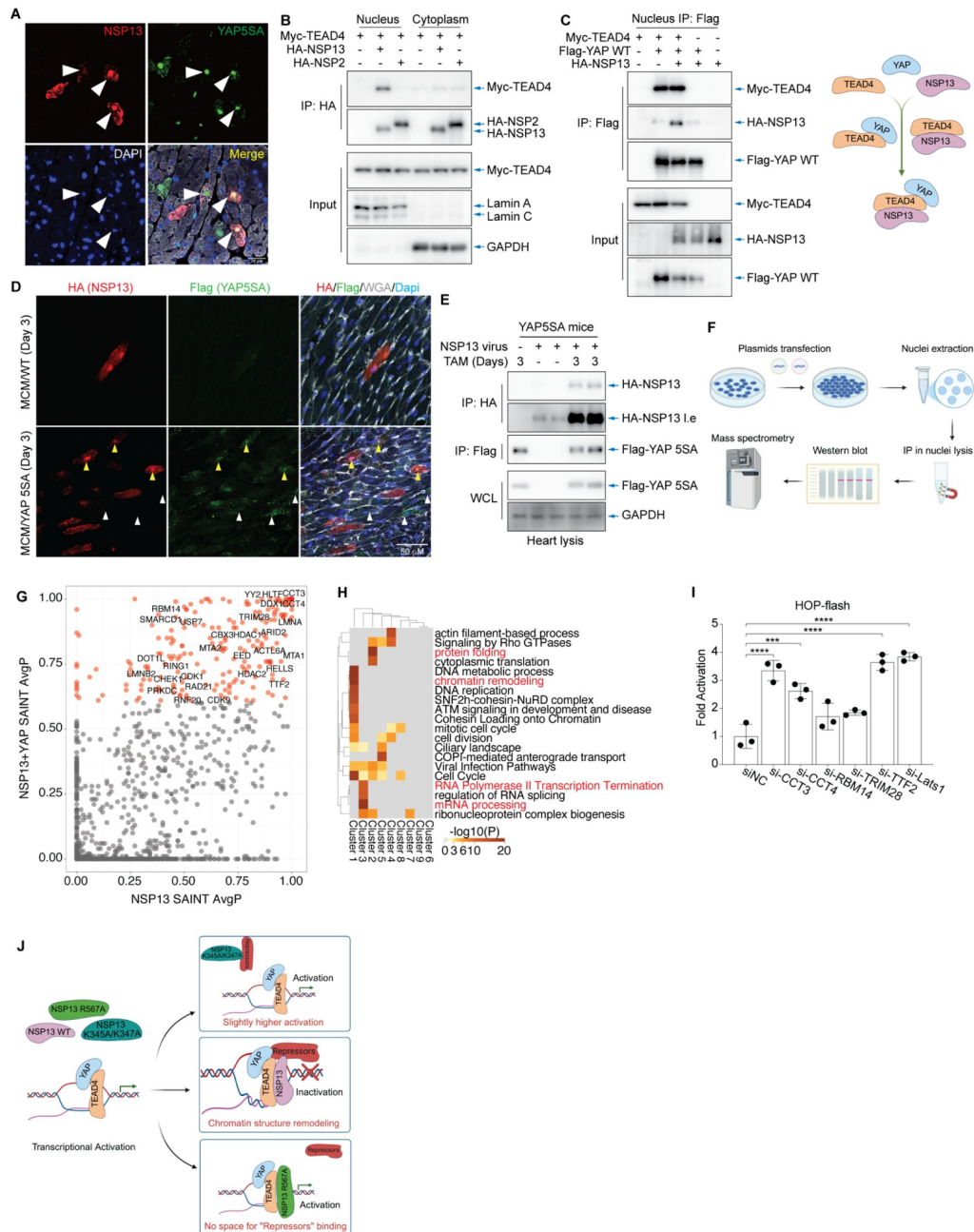


Figure 4.

NSP13 inactivates the YAP/TEAD4 complex by recruiting YAP repressors.

(A) Immunofluorescence imaging showing that NSP13 colocalized with YAP5SA in cardiomyocytes of YAP5SA transgenic mice 3 days after tamoxifen injection. (B) Co-IP: NSP13 interacts with TEAD4, a major binding partner of YAP, in the nucleus. (C) Co-IP of HEK293T nuclei: NSP13 does not disrupt the interaction between YAP and TEAD4, whereas TEAD4 promotes the interaction between YAP and NSP13. (D-E) Immunofluorescence imaging and western blot analysis reveal that NSP13 protein levels increase after YAP5SA expression is induced in YAP5SA mouse cardiomyocytes. (F) Workflow of IP-MS. (G) IP-MS in nuclei (IP: NSP13), suggesting NSP13 interacts with proteins with or without YAP co-expression. Significance Analysis of INteractome (SAINT), AvgP > 0.6 are labeled in red. (H) GO analysis in subclusters of NSP13 interacting proteins (SAINT, AvgP > 0.6, labelled with red in Figure S4C). (I) HOP-flash reporter assay: endogenous YAP activity is increased after the siRNA knockdown of CCT3 and TTF2 in HeLa cells. ***p < 0.001, ****p < 0.0001, one-way ANOVA. (J) Working model for NSP13 regulation of YAP/TEAD.

Supplement 5B [↗](#)), the candidate repressors exhibited relatively weak binding to NSP13, suggesting that NSP13 associates with these proteins indirectly, potentially through a larger multiprotein complex or via chromatin-dependent interactions (**Figure 4J** [↗](#)).

Next, we evaluated the functional consequence of siRNA knockdown of selected candidates in YAP transactivation (**Figure Supplement 5C** [↗](#)). Reporter assays revealed that knockdown of CCT3 and TTF2 increased endogenous YAP transactivation in HeLa cells (**Figure 4I** [↗](#) and **Figure Supplement 5D** [↗](#)). To confirm the function of CCT3, we performed quantitative PCR analysis of classical YAP target genes and observed increased expression of *Ctgf* and *Cyr61* following *Cct3* knockdown. These findings support a model where NSP13 suppresses YAP activity by recruiting suppressors to the YAP/TEAD4 complex.

Discussion

In this study, we observed that SARS-CoV-2 infection in human lung and hiPSC-CMs reduces YAP transcriptional activity, wherein the SARS-CoV-2 helicase NSP13 significantly inhibited YAP activity. Mechanistically, NSP13 directly interacts with TEAD4 to form a YAP/TEAD4/NSP13 complex in the nucleus, which recruits suppressors such as TTF2 and CCT3 to repress YAP-TEAD transcriptional activity (**Figure 4J** [↗](#)).

Our data reveal a helicase dependent inhibition of YAP by NSP13, where the K131, K345/K347, and R567 NSP13 residues are required. Based on published structural and biochemical studies, each of these residues uniquely supports helicase function: Substituting K131 with alanine (K131A) severely reduces helicase efficiency; K345/K347 are key DNA binding residues as mutating both (K345A/K347A) abolishes DNA binding; mutation of the ATP hydrolysis residue R567 (R567A) disables DNA unwinding. As illustrated in **Fig. 4J** [↗](#), NSP13 must bind DNA and hydrolyze ATP to unwind nucleic acids. This helicase dependent process likely enables NSP13 to remodel chromatin by binding TEAD and organizing YAP repressors at the YAP/TEAD complex to prevent YAP/TEAD transactivation. In support of this mechanism, the K345A/K347A mutant, unable to anchor to DNA, fails to repress YAP as YAP driven transcription is slightly increased in this mutant (**Figure 3B** [↗](#)). Likewise, the ATPase dead R567A can bind DNA but does not unwind and remodel chromatin to recruit YAP repressors, resulting in a loss of YAP suppression (**Figure 3B** [↗](#) and **3F** [↗](#)). Our model demonstrates that both DNA binding and ATP dependent unwinding are essential for NSP13 to suppress YAP transcriptional activity.

TEAD transcription factors, the major partners of YAP, are the primary nuclear effectors of Hippo-YAP signaling and play critical roles in cancer development ([11](#) [↗](#), [41](#) [↗](#)). We observed that NSP13 interacts with TEAD4, but does not disrupt the YAP/TEAD4 interaction, suggesting a novel regulatory mechanism for TEAD. IP-MS revealed that NSP13-interacting proteins such as TTF2 and CCT3 suppress the YAP-TEAD4 complex. TTF2, a SWI2/SNF2 family member, facilitates the removal of RNA polymerase II from the DNA template via ATP hydrolysis ([42](#) [↗](#), [43](#) [↗](#)). Importantly, termination of transcription complex elongation by TTF2 appears minimally affected by template position ([43](#) [↗](#)). CCT3 is a component of the chaperonin-containing T-complex, a molecular chaperone complex that promotes protein folding following ATP hydrolysis. Previous reports revealed that the liver cancer biomarker CCT3 ([44](#) [↗](#)) positively regulates YAP protein stability. We found that CCT3 interacts with NSP13 and inhibits YAP transactivation. Moreover, CCT3 knockdown increases YAP target gene expression, suggesting a context-dependent role of CCT3 in YAP regulation. While Co-IP assays detected weak interactions between NSP13 and CCT3 or TTF2, a strong NSP13/TEAD4 interaction was evident. These data suggest that NSP13 associates with CCT3 and TTF2 indirectly as part of a multiprotein complex in a chromatin structure-dependent manner, although further study is required to elucidate these mechanisms.

In conclusion, our study reveals a novel function of NSP13 in suppressing the YAP/TEAD transcriptional complex. These findings advance our understanding of the pathological effects of SARS-CoV-2 on host cells. Given that NSP13 directly interacts with TEAD4 and strongly suppresses YAP transactivation, this work provides a new avenue for modulating YAP activity as a potential treatment for YAP-driven diseases.

Materials and methods

Mice

We used α MyHC-MerCreMer mice, wild-type (WT) mice, and α MyHC-MerCreMer; YAP5SA mice (24 [24](#)). Mice had a mixed genetic background of C57BL/6 and 129SV. The AAV9 virus (a total of 5×10^{11} viral genomes, 120 μ l total volume) was delivered by retro-orbital injection two weeks before tamoxifen injection. For the survival experiment in **Figure 2** [2](#), two low doses of tamoxifen (10 μ g/g) were administered to 6-week-old mice by intraperitoneal injection. The mouse cardiac function was evaluated by echocardiography at day 4 and day 8 post-tamoxifen injection, and all the surviving mice were sacrificed at day 21. For the experiments in **Figure 4** [4](#), two tamoxifen doses were injected (**Figure 4A** [4](#), 10 μ g/g; **Figure 4D-E** [4](#), 50 μ g/g), and mouse hearts were collected at day three post-tamoxifen injection.

RNA-seq analysis

To analyze lung samples from COVID-19 patients, snRNA sequencing data of 7 control lungs and 35 COVID-19 lungs from GSE171668 (25 [25](#)) and GSE171524 (26 [26](#)) were downloaded and analyzed using the Seurat v4 software suite (45 [45](#)). A total of 223,106 nuclei were used in the analysis. Each sample was normalized and batch corrected using SCTransformation. Mitochondrial percentage was used to regress out technical variability between batches, and we used harmony (46 [46](#)) integration to remove technical variation among samples. The YAP score was evaluated using 38 canonical YAP target genes. For analyzing iPSC-CMs infected with SARS-CoV-2, bulk RNA-seq data were obtained from Perez-Bermejo et al. (23 [23](#)). With this dataset, we re-analyzed and evaluated the average expression of 302 cardiomyocyte-specific YAP target genes (24 [24](#)).

Expression plasmids

Expression plasmids encoding HA-tagged WT, mutant, or truncated NSP13 were generated by polymerase chain reaction (PCR) and subcloned into pXF4H (N-terminal HA tag) derived from pRK5 (Genetech). Myc-tagged WT or truncated TEAD4, flag-tagged WT YAP, flag-tagged YAP5SA, and Myc-tagged NSP13 interacting proteins, including CCT3, SMARCD1, EIF4A1, LMNA, TTF2, and YY2, were generated by PCR and subcloned into a pcDNA3 backbone. HA-tagged WT LATS2, LATS2 KR, and pRL-TK_Luc were gifts from Dr. Pinglong Xu's lab. Reporters of HOP-flash (#83467) and 8xGT10C-luciferase (#34615) were purchased from Addgene. The NSP plasmids used in YAP transactivation screening experiments were gifts from Dr. Nevan J. Krogan's lab. All plasmids were confirmed by performing DNA sequencing.

Cell culture and transfection

HEK293T and HeLa cells were cultured in Dulbecco's Modified Eagle Medium with 10% fetal bovine serum. Lipofectamine™ 2000 (ThermoFisher) or Lipofectamine™ 3000 (ThermoFisher) reagents were used for plasmid transfection. LipofectAmine RNAiMAX (ThermoFisher) was used for siRNA transfection.

Immunoprecipitation-mass spectrometry, database search, and data analysis

HEK293T cells were plated on a 150 mm tissue culture dish using 10% FBS DMEM 24 hours post-plating. Cells were transfected with Flag-YAP1, HA-NSP13, or Flag-YAP1+HA-NSP13 by Lipofectamine 3000 using standard manufacturer's protocols. Each experimental group was performed in duplicate to ensure reproducibility and reliability. 24 hours post-transfection, cells were collected in a hypotonic lysis buffer (5 mM NaCl, 20 mM HEPES, pH 7.5, 0.4% NP-40) supplemented with protease inhibitor (Roche) and phosphatase inhibitor (Roche). Cell suspensions were drawn into a 3 ml syringe using a 25G needle and expelled in one rapid stroke into a new tube. After three rounds of lysis, the suspension was centrifuged at 600 x g 4 °C for 10 minutes to pellet nuclei. The supernatant was collected in a different tube, and the nuclear fraction was washed in the buffer twice. After the final wash, nuclei were resuspended in 20 mM HEPES, 150 mM NaCl, 2.5 mM MgCl₂, 0.1% NP 40, and nuclear fractions were sonicated using Bioruptor® Pico for 10 cycles of 30 sec on and 30 sec off. Subsequently, lysate was centrifuged for 20 mins at 2 x 1000 g at 4°C. The supernatant was incubated with HA magnetic beads (ThermoFisher) for two hours at 4°C with continuous rotation. Antibody and protein-bound beads were washed thrice for 15 minutes in lysis buffer at 4 °C with constant rotation. Washed beads were boiled in 40 µL of 1X NUPAGE® LDS sample buffer (Invitrogen) and subjected to SDS-PAGE (NuPAGE 10% Bis-Tris Gel, Invitrogen). Eluted proteins were visualized with Coomassie Brilliant blue stain and excised into gel pieces according to molecular size. Individual gel pieces were destained and subject to in-gel digestion using trypsin (GenDepot T9600). Tryptic peptides were resuspended in 10 µl of loading solution (5% methanol containing 0.1% formic acid) and subjected to nanoflow LC-MS/MS analysis with a nano-LC 1000 system (Thermo Scientific) coupled to Orbitrap Elite (Thermo Scientific) mass spectrometer. Peptides were loaded onto a Reprosil-Pur Basic C18 (1.9 µm, Dr.Maisch GmbH, Germany) precolumn of 2 cm X 100 µm size. The precolumn was switched in-line with an in-house 50 mm x 150 µm analytical column packed with Reprosil-Pur Basic C18 equilibrated in 0.1% formic acid/water. Peptides were eluted using a 75 min discontinuous gradient of 4-26% acetonitrile/0.1% formic acid at a 700 nl/min flow rate. Eluted peptides were directly electro-sprayed into Orbitrap Elite mass spectrometer operated in the data-dependent acquisition mode acquiring fragmentation spectra of the top 25 strongest ions and under the direct control of Xcalibur software (Thermo Scientific).

MS/MS spectra were analyzed using the target-decoy mouse RefSeq database in the Proteome Discoverer 1.4 interface (Thermo Fisher), employing the Mascot algorithm (Mascot 2.4, Matrix Science). The precursor mass tolerance was set to 20 ppm, with a fragment mass tolerance of 0.5 daltons and a maximum of two allowed missed cleavages. Dynamic modifications included oxidation, protein N-terminal acetylation, and destreaking. Peptides identified from the Mascot results were validated at a 5% false discovery rate (FDR), and the iBAQ algorithm was applied to assess protein abundance, enabling comparisons of relative quantities among different proteins in the sample. The iBAQ value was determined by normalizing the summed peptide intensity by the number of theoretically observable tryptic peptides for each protein. PSM values from Proteome Discover were uploaded to the SAINT user interface on the CRAPOME website for SAINT analysis. SAINT probability and fold change were calculated using YAP IP as a user-defined control.

Luciferase reporter assay

HEK293T or HeLa cells were transfected with WT YAP-or YAP5SA-responsive HOP-flash plasmid or 8xGT10C-luciferase reporter, which has an open reading frame encoding firefly luciferase, along with the pRL-Luc with *Renilla* luciferase coding as the internal control for transfection and other expression vectors (NSPs), as specified. After 24 h of transfection with the indicated treatments, cells were lysed with passive lysis buffer (Promega). Luciferase assays were performed by using a dual-luciferase assay kit (Promega), and data were quantified with POLARstar Omega (BMG Labtech) and normalized to the internal *Renilla* luciferase control.

AAV9 viral packaging

Viral vectors were used as previously described (47 [↗](#)). The construct containing HA-tagged NSP13 was cloned into the pENN.AAV.cTNT, p1967-Q vector (AAV9-HA-NSP13). Empty vector-encoding green fluorescent protein was used as the control (AAV9-GFP). Both vectors were packaged into the muscle-trophic serotype AAV9 by the Intellectual and Developmental Disabilities Research Center Neuroconnectivity Core at Baylor College of Medicine. After being titered, viruses were aliquoted (1×10^{13} viral genome particles per tube), immediately frozen, and stored long-term at -80°C . Each aliquot was diluted in saline to make a 120- μl injection solution.

Echocardiography

For cardiac function analysis, echocardiography was performed on a VisualSonics Vevo 2100 system with a 550-s probe. B-mode images and M-mode images were captured on a short-axis projection. Ejection fraction, fractional shortening, diameter of diastolic left ventricle, diameter of systolic left ventricle, and end-diastolic volume were calculated by using a cardiac measurement package installed in the Vevo2100 system.

Histology and Immunofluorescence staining

Hearts were fixed in 4% paraformaldehyde overnight at 4°C , dehydrated in serial ethanol and xylene solutions, and embedded in paraffin. For immunofluorescence staining, slides were sectioned at 7- μm intervals. For paraffin sections, samples were deparaffinized and rehydrated, treated with 3% H_2O_2 in EtOH and then with antigen retrieval solution (Vector Laboratories Inc., Burlingame, CA, USA), blocked with 10% donkey serum in phosphate-buffered saline, and incubated with primary antibodies. Antibodies used were rabbit anti-HA (#3724, Cell signaling) and rat anti-flag (NBP1-06712, Novus Bio-logicals). Immunofluorescence-stained images were captured on a Zeiss LSM 780 NLO Confocal/2-photon microscope.

Statistical Analysis

For the data in our manuscript, the specific statistical test used is presented in the figure legend. In snRNA-seq analyses, Yap score differences between AT1 from Control and COVID-19 patients were identified via the Wilcoxon test. For analyzing iPSC-CMs infected with SARS-CoV-2, the expression score of cardiomyocyte YAP targets was evaluated by Student's t-test. Mouse survival rates in **Figure 2D** [↗](#) were analyzed by the log-rank (Mantel-Cox) test. Statistical significances were evaluated using three-way ANOVA and Šídák's multiple comparisons test for other cardiac data, including EF, FS, and systolic LV diameter. For reporter assays in cells and heart/body ratio in Figure S2E, the statistical significance of the observed differences in mean was evaluated using a one-way or two-way ANOVA and the post hoc Tukey's multiple comparisons test. P values less than 0.05 were considered statistically significant (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$).

Supplemental files

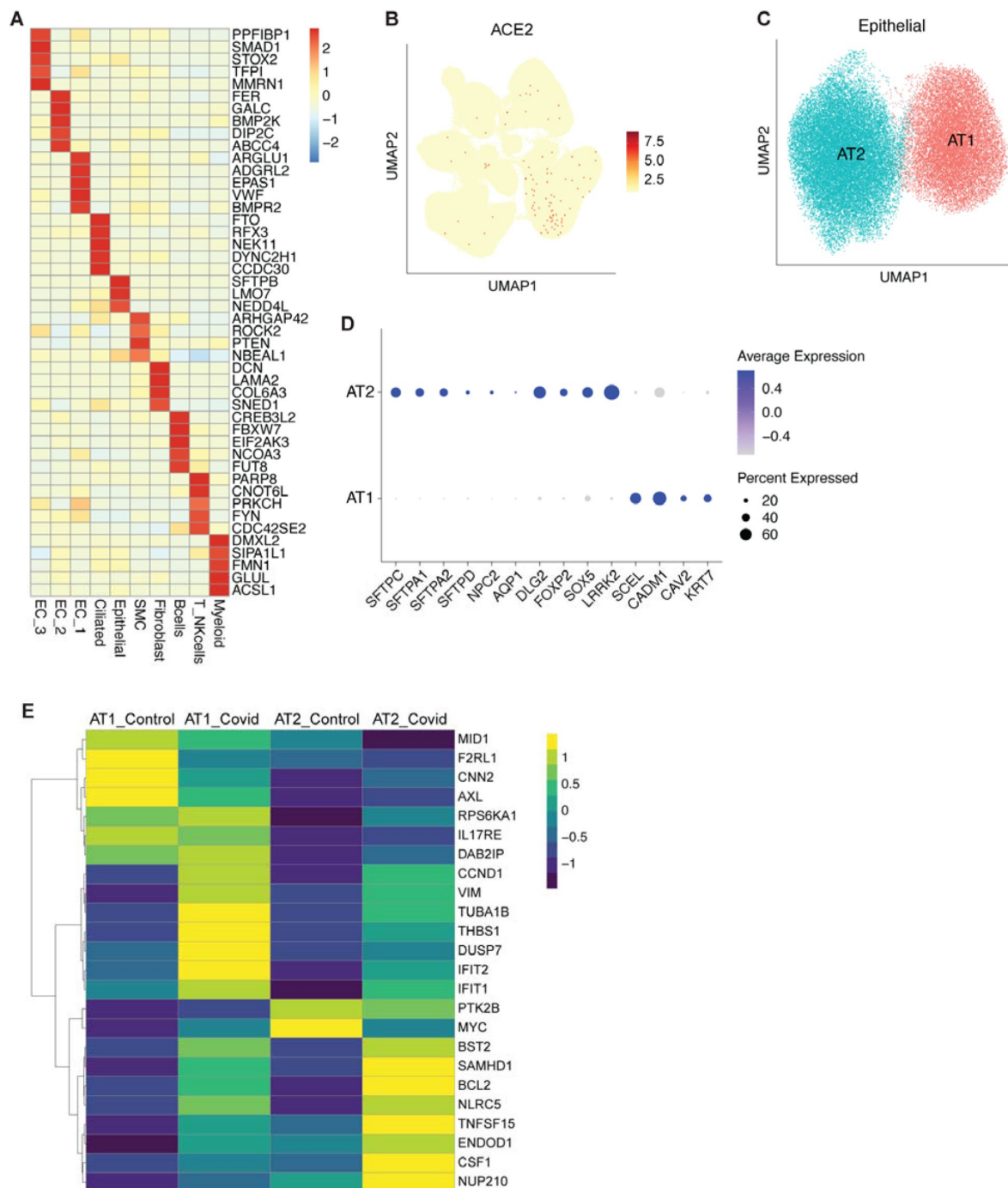


Figure supplement 1.

SARS-CoV-2 infection suppresses YAP activity *in vivo* and *in vitro*.

(A) Heatmap showing the most highly expressed markers for each cell type in human lung samples. (B) Uniform manifold approximation and projection (UMAP) visualization showing ACE2 expression in all cell types. (C) UMAP of epithelial cell subclusters. (D) Dot plot showing the relative expression of alveolar type 1 (AT1) and alveolar type 2 (AT2) markers in human lung samples. (E) Expression of Yap target genes involved in innate immune response or regulation in AT1 and AT2 cells from control and COVID-positive lung samples.

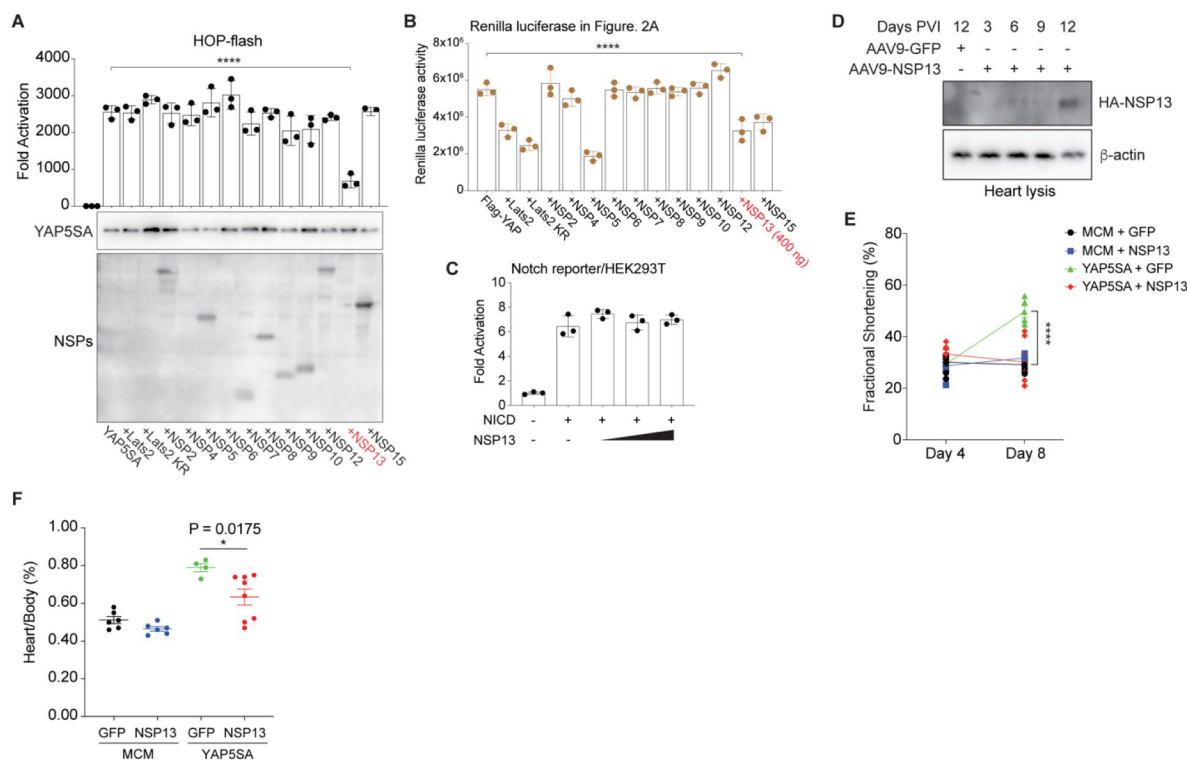


Figure supplement 2.

NSP13 inhibits YAP transactivation *in vitro* and *in vivo*.

(A) Reporter assay (HOP-flash) results indicating that NSP13 inhibited YAP5SA transactivation at low protein expression levels when compared with other NSPs. (n = 3 independent experiments; data are presented as mean ± SD). ****p<0.0001, one-way ANOVA. **(B)** Raw reads of Renilla luciferase in [Figure 2A](#). **(C)** Reporter assay (Notch reporter) results showing that NSP13 can not suppress NICD activation. (n = 3 independent experiments; data are presented as mean ± SD). **(D)** NSP13 expression at different time points in mouse hearts after AAV9-HA-NSP13 virus injection. **(E)** Fractional shortening was increased in YAP5SA mice at day 8 after tamoxifen injection (10 ug/g x2). This increase was reversed by NSP13 expression. ****p<0.0001, three-way ANOVA. **(F)** The overgrowth of the heart in YAP5SA mice, as evidenced by heart/body ratio, was reversed by NSP13 expression. *p=0.0175, one-way ANOVA.

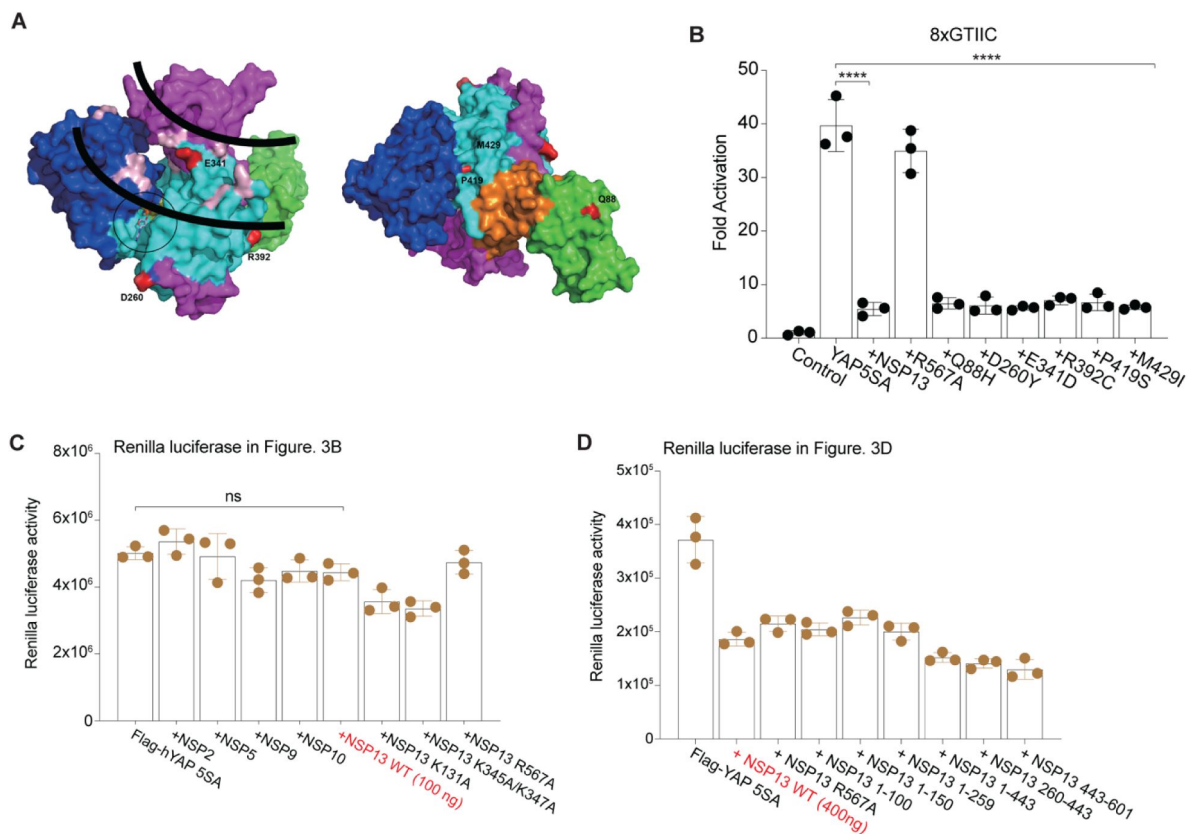


Figure supplement 3.

NSP13 helicase activity is required for suppressing YAP activity.

(A) The location of the NSP13 mutations from SARS-CoV2 variants on the NSP13 protein structure model (PDB: 7NN0). (B) Reporter assay (8xGTIIC) results indicating that NSP13 mutations did not affect its suppression on YAP5SA transactivation. ****p<0.0001, one-way ANOVA. (C) Raw reads of Renilla luciferase in Figure 3B. (D) Raw reads of Renilla luciferase in Figure 3D.

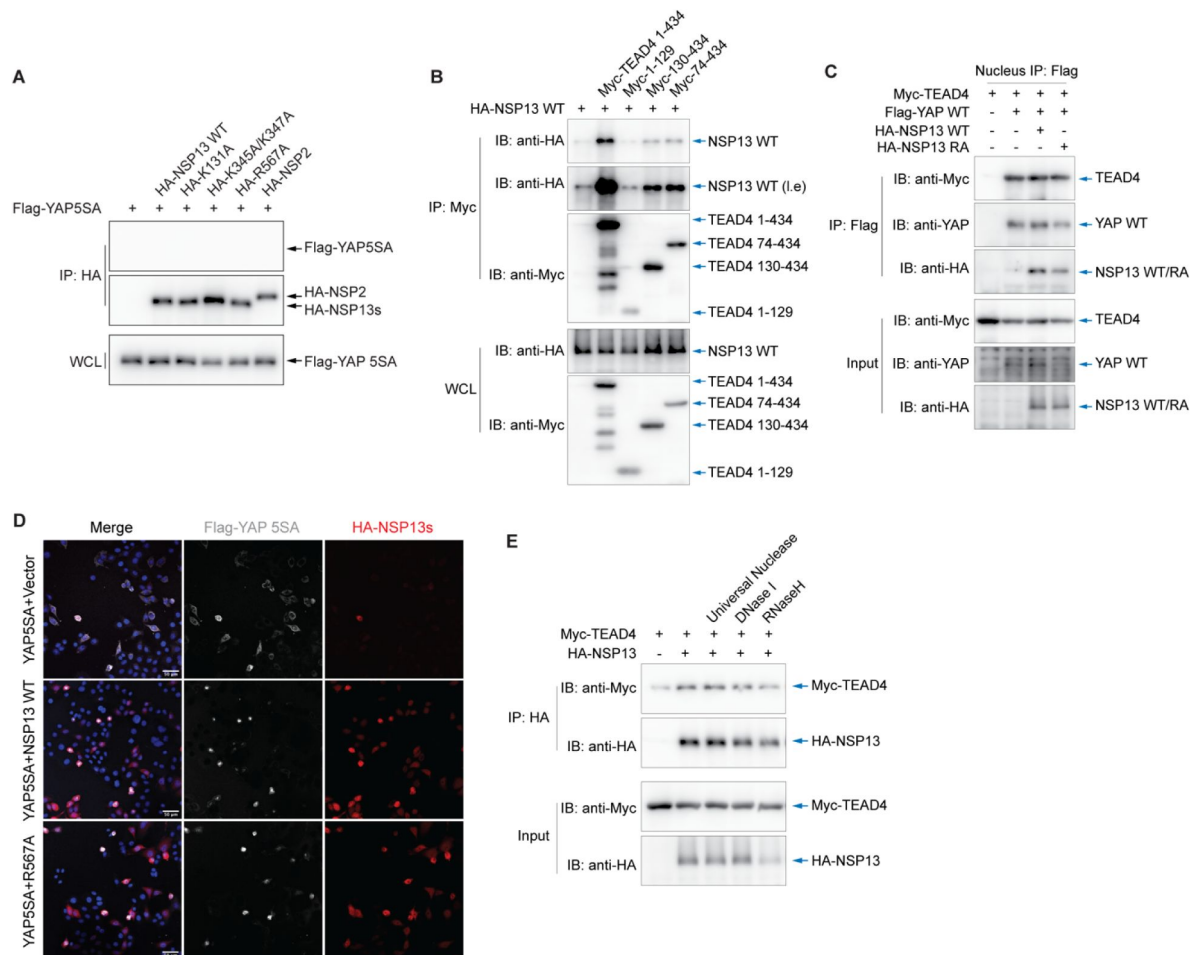


Figure supplement 4.

NSP13 interacts with TEAD4.

(A) Western blot analysis showed that neither wild-type nor mutant NSP13 directly interacted with YAP5SA in HEK 293T cells. (B) TEAD4 domain mapping experiments showing that both the N-terminus and C-terminus are required for the interaction with NSP13. (C) Co-immunoprecipitation in nucleus indicated that NSP13 wild-type and R567A did not disrupt YAP and TEAD4 interaction. (D) Immunofluorescence imaging showing that both NSP13 WT and R567A restricted YAP5SA in cell nucleus. (E) Western blot analysis of co-IP showed that nucleic acid removal did not disrupt the NSP13–TEAD4 interaction.

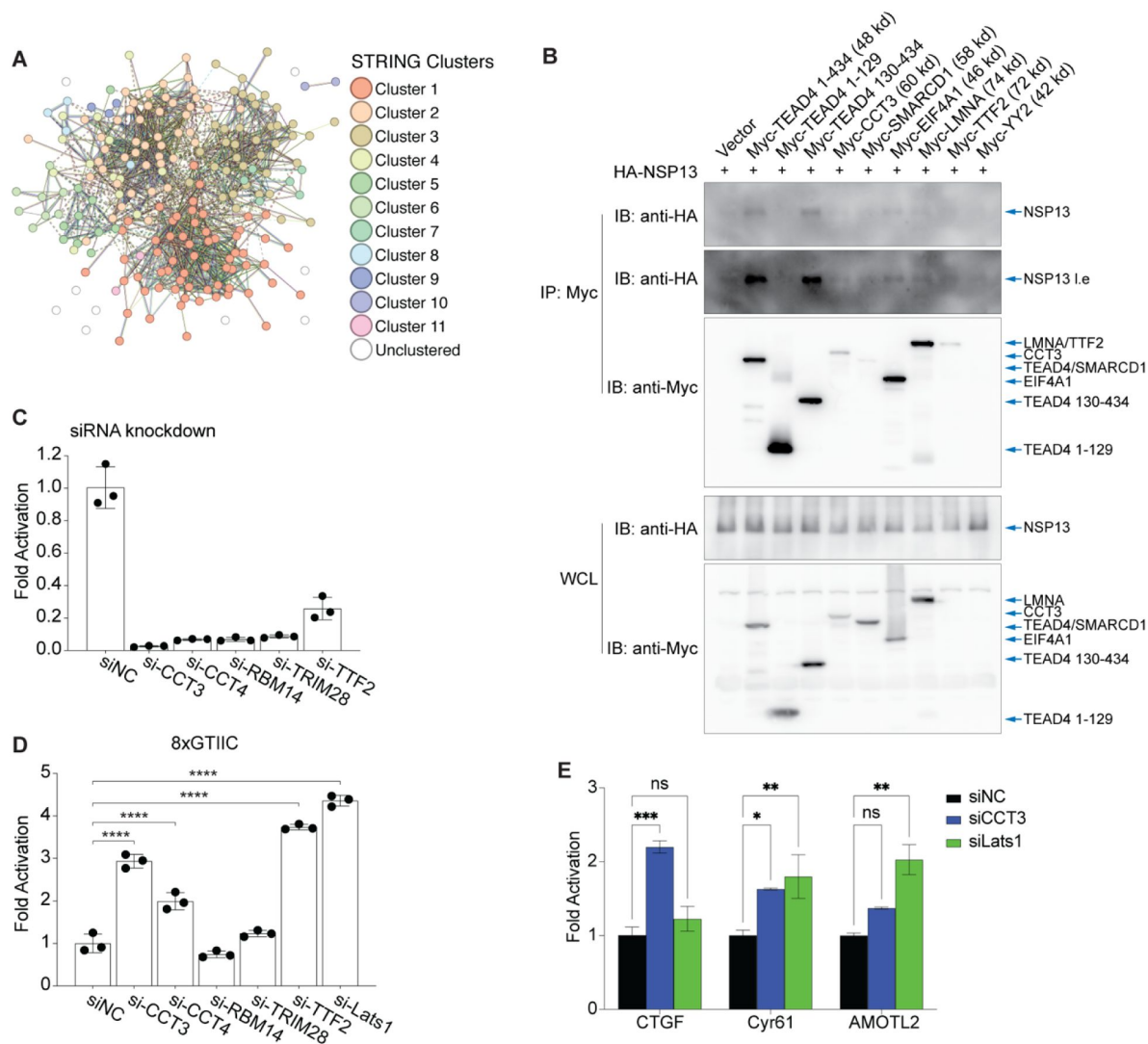


Figure supplement 5.

NSP13 inactivates the YAP/TEAD4 complex by recruiting YAP repressors.

(A) Immunoprecipitation mass spectrometry in nuclei (IP: NSP13) suggesting NSP13 interacting proteins with or without YAP co-expression. Significance Analysis of INteractome (SAINT), AvgP >0.6 were labeled with red. (B) co-IP assays indicated that the NSP13 interactors from mass spectrometry had weaker binding with NSP13 compared to TEAD4. (C) siRNA knockdown efficiency for genes in HeLa cells, determined by using quantitative polymerase chain reaction. (D) Reporter assay (8xGTIIC) results in HeLa cells revealed that endogenous YAP activity was increased after the siRNA-mediated knockdown of CCT3 and TTF2. ****p<0.0001, one-way ANOVA. (E) Quantitative PCR analysis indicated increased expression of *Ctgf* and *Cyr61* following *Cct3* knockdown. *p<0.05, **p<0.01, ***p<0.001, two-way ANOVA.

Data and materials availability

All data are available in the main text or the supplementary materials.

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Additional files

Supporting data values [↗](#)

WB raw figure [↗](#)

sp1 IP-MS raw data [↗](#)

sp2 IP-MS analysis [↗](#)

sp3 STRING analysis [↗](#)

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Reviewer #1 (Public review):

In the revised manuscript, Meng et al. report that SARS-CoV-2 infection suppresses YAP target gene transcription in both patient lung samples and iPSC-derived cardiomyocytes. Among the tested viral proteins, the helicase nonstructural protein 13 (NSP13) was identified as a key factor that impairs YAP/TEAD transcriptional activity. Through mutagenesis and protein-protein interaction studies, the authors propose a mechanism where NSP13 binds YAP/TEAD complex, remodels chromatin structure, and recruits transcriptional repressors to inhibit YAP/TEAD's transcriptional activity.

Overall, this study uncovers a novel regulation of Hippo signaling by SARS-CoV-2 through NSP13, suggesting a potential role of this growth-related pathway in host innate immune response to viral infection. While these findings are intriguing, future studies are needed to validate the involvement of YAP/TEAD in patient tissues and to assess their potential as therapeutic targets against SARS-CoV-2.

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Reviewer #2 (Public review):**Summary:**

The manuscript by Meng et al. describes a role for the coronavirus helicase NSP13 in the regulation of YAP-TEAD-mediated transcription. The authors present data that NSP13 expression in cells reduces YAP-induced TEAD luciferase reporter activity and that NSP13 transduction in cardiomyocytes blocks hyperactive YAP-mutant phenotypes in vivo. Mechanisms by which viral proteins (particularly those from coronaviruses) intersect with cellular signaling events is an important research topic, and the intersection of NSP13 with YAP-TEAD transcriptional activity (independent of upstream Hippo pathway mediated signals) offers new knowledge that is of interest to a broad range of researchers.

Strengths:

The manuscript presents convincing data mapping the effects of NSP13 on YAP-TEAD reporter activity to the helicase domain. Moreover, the in vivo data demonstrating that NSP13 expression in YAP5SA mouse cardiomyocytes increased survival animal rates, and restored cardiac function is striking and is supportive of the model presented.

Weaknesses:

While there are some hints at the mechanisms by which NSP13 regulates YAP-TEAD activity through the identification of NSP13-associated proteins by mass spec, the relationships and functions of these factors in the context of YAP-TEAD regulation requires further study in the future.

<https://doi.org/10.7554/eLife.100248.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):*Major points*

(1) The authors discovered a novel regulation of the Hippo-YAP pathway by SARS-CoV-2 infection but did not address the pathological significance of this finding. It remains unclear why YAP downstream gene transcription needs to be inhibited in response to SARS-CoV-2 infection. Is this inhibition crucial for the innate immune response to SARS-CoV-2? The authors should re-analyze their snRNA-seq and bulk RNA-seq data described in Figure 1 to determine whether any of the affected YAP downstream genes are involved in this process.

We appreciate the reviewer's suggestion to clarify the pathological significance of YAP pathway inhibition in SARS-CoV-2 infection. To address this, we re-analyzed our snRNA-seq and bulk RNA-seq datasets to determine whether YAP target genes overlap with known mediators of the innate immune response. As described in Fig. 1C, bulk RNA-seq revealed decreased expression of multiple YAP downstream targets linked to innate immune regulation (e.g., *Thbs1*, *Ccl2*, *Axl*, and *Csf1*) in SARS-CoV-2-infected cells in vitro.

snRNA-seq of alveolar type I (AT1) cells from COVID-19 patients revealed a more complex landscape: While we observed reduced YAP activity overall (Fig. 1G), multiple YAP target genes involved in innate immunity and cytokine signaling were paradoxically elevated (Supplemental Fig. 1E). Several factors likely explain these conflicting observations: 1. In the lung, AT1 cells (which are critical for gas exchange) may cell specifically respond to virus infection by upregulating genes related to immune response by other signaling pathway(s); 2. In vivo, SARS-CoV-2 infection triggers a surge in cytokines, chemokines, and other local factors that can differentially modulate YAP binding sites and thus affect its downstream targets, a complexity not fully captured in vitro; 3. YAP is highly sensitive to mechanical signals and tissue architecture. The 3D structure of altered cell-cell junctions in infected lung tissue, and fluid shear stress in the alveolar space could shape YAP target gene transcription differently from simplified monolayer cell cultures.

We have expanded the results section of the new version to include the above points. We also acknowledge that ongoing and future work is needed to delineate the exact molecular and tissue-specific pathways through which YAP inhibition confers a potential advantage in combating SARS-CoV-2.

(2) The authors concluded that helicase activity is required for NSP13-induced inhibition of YAP transcriptional activity based on mutation studies (Figure 3B). This finding is somewhat confusing, as K131, K345/K347, and R567 are all essential residues for NSP13 helicase activity while mutating K131 did not affect NSP13's ability to inhibit YAP (Figure 3B). Additionally, there are no data showing exactly how NSP13 inhibits the YAP/TEAD complex through its helicase function. This point was also not reflected in their proposed working model (Figure 4H).

We appreciate the reviewer's concerns regarding the helicase-dependent inhibition of YAP by NSP13, particularly the roles of K131, K345/K347, and R567. Based on published structural and biochemical studies, each of these residues uniquely supports helicase function (1): K131 is crucial for stabilizing the NSP13 stalk region by interacting with S424. Substituting K131 with alanine (K131A) reduces helicase efficiency but does not completely abolish it; K345/K347 are key DNA-binding residues, and mutating both (K345A/K347A) largely prevents NSP13 from binding DNA, thus eliminating unwinding. R567 is critical for ATP hydrolysis, and the R567A mutant retains DNA binding capacity but fails to unwind it. In Fig. 3B, K131A suppresses YAP transactivation to nearly the same extent as wild-type NSP13, suggesting that partial helicase activity is sufficient for complete YAP/TEAD inhibition. Conversely, the K345A/K347A and R567A mutants show markedly diminished repression, underscoring the importance of DNA binding and ATP hydrolysis.

As the new Fig. 4J illustrates, NSP13 must bind DNA and hydrolyze ATP to unwind nucleic acids. This helicase-dependent process likely enables NSP13 to remodel chromatin structure by binding TEAD and properly organize YAP repressors at YAP/TEAD complex to prevent YAP/TEAD transactivation. In support of this mechanism, the K345A/K347A mutant, unable to anchor to DNA, fails to repress YAP and slightly increases YAP-driven transcription (Fig. 3B), presumably by mislocalizing YAP repressors. Likewise, the ATPase-dead R567A can bind DNA but does not unwind and remodel chromatin to recruit YAP repressors, resulting in a loss of YAP suppression (Fig. 3B and 3F). Our revised model demonstrates that both DNA binding and ATP-dependent unwinding are essential for NSP13 to suppress YAP transcriptional activity. We have updated the results, discussion, and model accordingly.

(3) The proposed model that NSP13 binds TEAD4 to recruit repressor proteins and inhibits YAP/TEAD downstream gene transcription (Figure 4H) needs further characterization. Second, NSP13 is a DNA-binding protein, and its nucleic acid-binding mutant K345A/K347A failed to inhibit YAP transcriptional activity (Figure 3B). The authors should investigate whether NSP13 could bind to the TEAD binding sequence or the nearby sequence on the genome to modulate TEAD's DNA binding ability. Third, regarding the identified nuclear repressors, the authors should validate the interaction of NSP13 with the ones whose loss activates YAP transcriptional activity (Figure 4G). Lastly, why can't NSP13 bind TEAD4 in the cytoplasmic fractionation if both NSP13 and TEAD4 are detected there (Figure 3B)? This finding indicates their interaction is not a direct protein-protein interaction but is mediated by something in the nucleus, such as genomic DNA.

(1) Low TEAD expression in HEK293T cells: Our IP-MS experiments were performed in HEK293T cells, which, according to the Human Protein Atlas, express TEAD1–4 at comparatively low levels (TEAD1: 16.5, TEAD2: 16.4, TEAD3: 4.9, TEAD4: 38.7 nTPM). In contrast, HeLa cells, where we successfully validated NSP13-mediated YAP suppression (Fig. 4H, Supplementary Fig.5B–D), show higher expression of these TEAD isoforms (TEAD1: 97.1, TEAD2: 27.3, TEAD3: 12.2, TEAD4: 48.1 nTPM). Therefore, insufficient TEAD abundance in HEK293T cells may limit the sensitivity needed to detect TEAD–NSP13 interactions in our proteomic screens.

(2) Transience and potential DNA dependence: Our co-immunoprecipitation (co-IP) experiments (Fig. 4B, Supplementary Fig.4C–E) indicated that NSP13–TEAD4 binding is low-affinity. Under standard IP-MS conditions (which typically do not include chemical cross-linkers or nucleic acids to stabilize transient complexes), weak or short-lived interactions can be lost during washes or sample processing.

(3). Additional supporting evidence: We carefully checked our IP-MS data and found that the well-known TEAD binding proteins, including CTBP1/2 and GATA4, were pulled down, suggesting TEAD's absence does not rule out an NSP13–TEAD association.

(3a) We acknowledge that our NSP13 immunoprecipitation–mass spectrometry (IP-MS) did not identify any TEAD proteins (Fig. 4G and IP-MS tables). Several factors likely contributed to this outcome:

(3b) We sincerely appreciate the reviewer's insightful suggestion. While we agree that mapping NSP13 occupancy at individual TEAD-binding motifs is valuable, we respectfully consider this to be beyond the scope of the current study. Biochemical and structural work on coronavirus NSP13 shows that it recognizes nucleic-acid substrates primarily through their 5' single-stranded overhang and duplex architecture, not through a defined base sequence(2, 3). Accordingly, our data (Fig. 3B and 3F) indicate that DNA binding ability, rather than recognition of a specific motif, enables NSP13 to perform its helicase activity in proximity to TEAD and recruit repressors. Moreover, the DNA-binding mutant K345A/K347A and the

ATPase-dead mutant R567A both fail to suppress YAP/TEAD transcription despite retaining the ability to interact with TEAD (Fig. 3B). These loss-of-function phenotypes demonstrate that NSP13's chromatin engagement and unwinding activity, rather than sequence-restricted targeting, are essential for repression. For these reasons, motif-specific binding assays were not pursued in this revision, but we clarified in the discussion that NSP13's DNA engagement is likely structural or TEAD-dependent, rather than sequence-directed. We also highlighted this as an important avenue for future investigation.

(3c) To validate the NSP13 interacting proteins from our IP-MS data, we generated plasmids expressing several candidates (CCT3, SMARCD1, EIF4A1, LMNA, TTF2, and YY2) and performed co-IP assays. As predicted, we confirmed the robust interaction between NSP13 and TEAD (Supplemental Fig. 5E). However, these putative nuclear repressors exhibited weak binding to NSP13 compared with TEAD4, suggesting that NSP13 associates with them indirectly, possibly as part of a larger multiprotein complex or depending on the chromatin structure, rather than via direct protein-protein interaction (Fig. 4J).

(3d) We appreciate the reviewer's question. To investigate whether their association might be DNA-dependent, we performed co-IP experiments using nuclear lysates in the presence or absence of various nucleases: Universal Nuclease (which degrades all forms of DNA and RNA), DNase I (which cleaves both single- and double-stranded DNA), and RNase H (which selectively cleaves the RNA strand in RNA/DNA hybrids). Our findings revealed that nucleic acid removal did not disrupt the NSP13/TEAD4 interaction (Supplemental Fig. 4E), indicating that their binding is not solely mediated by DNA or RNA.

Reviewer #2 (Public Review):

Specific comments and suggestions for improvement of the manuscript:

(1) NSP13 has been reported to block, in a helicase-dependent manner, episomal DNA transcription (PMID: 37347173), raising questions about the effects observed on the data shown from the HOP-Flash and 8xGTIIIC assays. It would be valuable to demonstrate the specificity of the proposed effect of NSP13 on TEAD activation by YAP (versus broad effects on reporter assays) and also to show that NSP13 reduces the function of endogenous YAP-TEAD transcriptional activity (i.e., does ectopic NSP13 expression reduce the expression of YAP induced TEAD target genes in cells).

We appreciate the reviewer's comments and have carefully revisited the conclusions from the published paper(4) (PMID: 37347173), which reported that NSP13 suppresses episomal DNA transcription, as evidenced by reduced Renilla luciferase (driven by the herpes simplex virus thymidine kinase promoter) and GFP expression upon co-expression with NSP13. For our experiments, we used a dual-luciferase assay with Renilla luciferase (under the same promoter) as an internal control. After re-examining our raw Renilla luciferase data (now provided in the supplemental Excel file "Supporting data value"), we found that while 100 ng of NSP13 did not affect Renilla luciferase levels, 400 ng of NSP13 reduced them by approximately 50% relative to the YAP5SA-only group (Supplemental Fig. 2B, Fig. 3C-D). We observed a similar reduction with NSP13 truncation mutants—an outcome not fully consistent with the published study (Supplemental Fig. 3D, PMID: 37347173). However, unlike their finding of robust episomal DNA suppression, our data indicate that the K345A/K347A mutant of NSP13, which lacks DNA-binding ability, completely lost its suppressive effect (Fig. 3B).

We performed additional Notch reporter assays to address the concern that NSP13 might nonspecifically inhibit episomal DNA transcription (including the HOP-Flash and 8xGTIIIC reporters). These experiments revealed that co-expression of NSP13 with NICD (Notch intracellular domain) does not suppress Notch signaling (Supplemental Fig. 2C), indicating that NSP13 does not globally block all reporter systems. To evaluate whether NSP13 reduces

endogenous YAP-TEAD activity, we transiently overexpressed NSP13 WT and its R567A mutant in HeLa cells. However, bulk RNA-seq and qPCR analyses did not reveal a clear decrease in YAP target genes, possibly due to the low transfection efficiency (< 50%, Supplemental Fig.4D). Interestingly, we observed that YAP5SA was predominantly retained in the nucleus upon NSP13 or R567A co-expression, suggesting that NSP13 (or together with its interacting partners) restricts YAP5SA cytoplasmic shuttling. Future studies will involve stable cell lines expressing NSP13 WT or R567A to better characterize the mechanisms driving YAP5SA nuclear retention and clarify how NSP13 specifically suppresses YAP activity.

(2) While the IP-MS experiment may have revealed new regulators of TEAD activity, the data presented are preliminary and inconclusive. No interactions are validated and beyond slight changes in TEAD reporter activity following knockdown, no direct links to YAP-TEAD are demonstrated, and no link to NPS13 was shown. Also, no details are provided about the methods used for the IP-MS experiment, raising some concerns about potential false positive associations within the data.

We appreciate the reviewer's feedback regarding our IP-MS findings and acknowledge that additional validation is required to establish definitive links between the identified putative regulators, YAP-TEAD, and NSP13. We have taken the following steps (and plan further experiments) to address these concerns:

(2a) Co-IP validation: Same with the answer for Reviewer #1 (3c), we generated plasmids expressing several top candidate interactors from the IP-MS data (CCT3, SMARCD1, EIF4A1, LMNA, TTF2, and YY2) and performed direct co-IP assays in a more controlled setting. The results indicated that these putative NSP13 interactors had weaker binding compared to TEAD4, implying that NSP13 may associate with them as part of a larger complex or depending on the chromatin structure rather than through a direct protein-protein interaction (Fig. 4J).

(2b) qPCR validation: Beyond reporter assays for evaluating YAP transactivation after the candidate YAP suppressor knockdown (Fig. 4H and Supplemental Fig. 5C), we performed qPCR to detect YAP activation on endogenous YAP-TEAD target genes (e.g., CTGF, CYR61, and AMOTL2) after CCT3 knockdown. Expression of CTGF and CYR61 was higher compared to control (Supplemental Fig. 5D), strengthening the case for an interaction relevant to YAP-TEAD signaling.

(2c) To investigate how NSP13-interacting proteins link to the YAP/TEAD complex, we examined the IP-MS dataset and identified several well-known YAP and TEAD binding partners, including CTBP1/2 (TEAD-binding), GATA4 (TEAD-binding), and multiple 14-3-3 isoforms (YWHAZ/YWHAB/YWHAH/YWHAQ, YAP binding). These findings suggest that NSP13 may form a larger nuclear complex with YAP/TEAD and associated cofactors. In the future, we will determine whether these putative TEAD regulators also interact with NSP13 under various conditions (e.g., in the presence or absence of DNA) and whether co-expression of NSP13 influences their association with YAP or TEAD. This approach will clarify how NSP13 might leverage these factors to regulate YAP-TEAD function.

(2e) For the mass spectrometry experiments, HEK293T cells were transfected with Flag-YAP1, HA-NSP13, or Flag-YAP1 + HA-NSP13 according to the manufacturer's standard protocols. After nuclear extraction and lysis, the supernatant was incubated with HA magnetic beads to immunoprecipitate (IP) NSP13. The IP samples were subsequently analyzed by mass spectrometry to identify NSP13-associated proteins (Fig. 4F). Each experimental condition was performed in duplicate to ensure reproducibility. We included an appropriate negative control (Flag-YAP1) and stringent data-filtering criteria to minimize false positives. We apologize for not including these details in our original Methods section; in this revised

manuscript, we have fully described the number of replicates, the controls used, and our data analysis pipelines.

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